Year 12 Maths - Objects in contact

Similar questions - Friction, inclined planes and connected particles

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A block B of mass 2 kg lies on a rough horizontal table. It is attached by a light inextensible string, passing over a smooth pulley at the edge of the table, to a particle P of mass 3 kg which hangs freely. The system is released from rest and P accelerates downwards at $\frac{1}{2}g$.

(a) Find the tension in the string. **[3]**

(b) Find the coefficient of friction between B and the table. **[3]**

(c) P hits the floor after falling 1.2 m and does not rebound. Assuming B does not reach the pulley, find the further distance B travels before it comes to rest. **[4]**

2 Particle A of mass 4 kg lies on a rough plane inclined at 30° to the horizontal. It is attached by a light inextensible string, passing over a smooth pulley at the top of the plane, to a particle B of mass 3 kg that hangs freely. The string is parallel to the line of greatest slope.

(a) When the system is on the point of moving, with A about to slide up the plane, show that the coefficient of friction is $\frac{\sqrt{3}}{6}$ and find the tension. **[6]**

(b) Instead, the coefficient of friction is $\frac{\sqrt{3}}{9}$ and the system is released from rest. Find the acceleration of A and the tension in the string. **[6]**

(c) After 2 seconds the string breaks. Find the further distance that A travels up the plane. **[4]**

Question 1

(a) P: $3g - T = 3\left(\frac{1}{2}g\right)$, so $T = 1.5g = 14.7$ N

(b) B: $T - F = 2\left(\frac{1}{2}g\right)$, so $F = 14.7 - 9.8 = 4.9$ N. $R = 2g = 19.6$ N, so $\mu = \dfrac{4.9}{19.6} = 0.25$

(c) $v^2 = 2\left(\frac{1}{2}g\right)(1.2) = 11.76$. After the string goes slack B decelerates at $\mu g = 2.45\ \text{m s}^{-2}$. Distance $= \dfrac{11.76}{2(2.45)} = 2.4$ m

Question 2

(a) B: $T = 3g$. A (limiting, up the plane): $T = 4g\sin 30^\circ + \mu R$ and $R = 4g\cos 30^\circ$. $3g = 2g + \mu \cdot 4g \cdot \frac{\sqrt{3}}{2}$, so $\mu = \dfrac{1}{2\sqrt{3}} = \dfrac{\sqrt{3}}{6}$. $T = 3g = 29.4$ N

(b) $\mu R = \dfrac{\sqrt{3}}{9} \times 4g \times \dfrac{\sqrt{3}}{2} = \dfrac{2g}{3}$. B: $3g - T = 3a$. A: $T - 2g - \dfrac{2g}{3} = 4a$. Adding: $3g - 2g - \dfrac{2g}{3} = 7a$, so $a = \dfrac{g}{21} = 0.467\ \text{m s}^{-2}$. $T = 3(g - a) = 28$ N

(c) At $t = 2$: $v = 2a = 0.933\ \text{m s}^{-1}$. A decelerates at $g\sin 30^\circ + \mu g\cos 30^\circ = 4.9 + 1.633 = 6.533\ \text{m s}^{-2}$. Distance $= \dfrac{0.933^2}{2(6.533)} = 0.0667$ m (6.67 cm)